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Financial Strain and Emergency Department Use: An Analysis of Patient Journeys

Health systems typically see patient visits as isolated snapshots, but those visits tell a much larger story. A patient who shows up to the emergency room multiple times in a single year is not just dealing with a medical issue. There is often something happening outside the hospital that is driving them there. Our team set out to explore whether financial strain, as captured through social determinants of health screening questions already administered at intake, could predict emergency department utilization among patients at Storrs Valley Hospital. The central question was straightforward: are patients facing financial hardship being pushed into the ER not because they are sicker, but because they lack the resources to access routine care?

SVH administers a broad set of SDOH screening questions across twelve domains, including alcohol use, depression, food insecurity, housing stability, intimate partner violence, physical activity, social connections, stress, transportation needs, and financial resource strain. From these, we selected questions that fell under the umbrella of financial and material hardship. We identified five domains relevant to our research question: utility shutoff threats, rent or mortgage instability, food insecurity, homelessness, and transportation barriers. Before fitting our model, we examined the correlation structure among the original screening questions. Two pairs were highly collinear. The two food insecurity questions correlated at $r = 0.86$, and the two transportation questions correlated at $r = 0.84$. Keeping both members of each pair in the model would have inflated their individual coefficients, so we collapsed each pair into a single binary indicator. This left us with five clean predictors, each coded as 0 or 1.

We then fit a logistic regression model on 32,079 patients, predicting whether each patient had any emergency department visit. All five domains were positively associated with ER use. Transportation barriers emerged as the strongest predictor with a coefficient of 0.390, followed by utility shutoffs at 0.282, rent strain at 0.133, homelessness at 0.119, and food insecurity at 0.060. To visualize the cumulative effect, we constructed a composite burden score by summing each patient's binary indicators across the five domains, producing a score from 0 to 5. Patients with a score of zero had an ER visit rate of roughly 41 percent. At a score of one, that rate climbed to about 55 percent. By a score of three it reached 65 percent, and patients who flagged positive on all five domains had an ER visit rate of approximately 77 percent. The relationship was monotonically increasing at every level, with no exceptions.

The finding that transportation is the single strongest predictor carries significant practical weight. A patient who cannot get to a clinic does not stop being sick. They wait. They miss follow-up appointments. Prescriptions go unfilled. Conditions that could have been managed in a routine visit deteriorate until the only option left is the emergency department, which is the most expensive and least effective form of care in the system. This is not a medical failure. It is a logistics problem, and it is one that SVH already has the data to address. The hospital screens every patient for transportation barriers at intake. The next step is acting on those answers: flagging patients with high burden scores and connecting them with transportation assistance, food programs, and social work referrals before they end up back in the

ER. The infrastructure to identify these patients already exists. What remains is the decision to intervene.